Retrieval Augmented Generation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach a source-supported answer and an honest gap.

Supply complete passages, preserve conditional facts and separate the retained chunking/ranking strands.

Keep one habit: Ask which exact passage supports each claim and what the selected context does not answer.

MAIN LESSON · SECTION 1

Prepare the main lesson and its retained extension

> Use fictional records and schedule separate extension time.

Goal and preparation

Primary target: Explain the **mechanism** and terminology of building a **grounded answer**

Two 45-minute sessions. A **grounded answer** connects each factual claim to **evidence** actually retrieved for the task. Retrieval success does not automatically make generation faithful.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can explain retrieval augmented generation using a traced example and a stated limitation.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Main plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra

> explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval and readiness

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of retrieval and indexing

Reference example, not the only acceptable wording: Teacher explanation: Retrieval returns stored records selected for a **query**.

An index is a searchable representation of a versioned corpus; generated wording is a later operation. Assign stable document IDs, versions and permitted access before indexing. Retrieve only authorised records.

Preserve source references and define behaviour when a document is missing, outdated or access-restricted.

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of conflicting and time-sensitive sources

Reference example, not the only acceptable wording: Teacher explanation: Changing facts require a stated time and source-version policy. Current-looking records can conflict without enough **evidence** to choose between them.

For this toy task a signed amendment supersedes the referenced original from its effective time. Keep both records. A newer timestamp alone has no authority unless the policy grants it.

Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

Reference example, not the only acceptable wording: Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response.

Errors at one stage can propagate even when later stages run correctly. Track source modality, segment/time references, transcription, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ Materials and extension planning

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Apply the connected idea: Chunking and document structure

Two 45-minute sessions. **Chunk**ing divides documents into retrievable units. A useful **chunk** retains enough structure to interpret a passage, including headers and exceptions.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Apply the connected idea: Lexical and vector retrieval

Two 45-minute sessions. Lexical retrieval matches terms; **vector retrieval** compares numerical representations. Each can fail when the representation misses the relevant distinction.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Model Memory vs Retrieval

Source Quality

Speech and Audio Systems

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Sources: A='Library opens 9'; B='Cafe opens 8'; C='Library closes 5'. A question asks when the library opens. The **retriever** returns only B and C. What should the generator do?

Concept to explain

Week 18: Retrieval-Augmented Generation

Term	Meaning
query	A question or search request used to obtain information.
retriever	The component that searches for and selects source material relevant to a query.
relevance	How directly information helps answer the stated question or perform the task.
grounded answer	An answer whose factual claims are supported by identified evidence relevant to the question.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach retrieval and faithful claim use as separate stages

> Keep source selection, wording and checking visible.

Model the simple evidence route

Note A says library opens at 9. Note B says returns go in the blue box.

Answer opening time with **evidence**.

Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

Stage	Evidence to preserve
Question	Entity, requested fact and relevant time
Retrieve	Eligible passage IDs, versions and scope
Inspect	Direct support, linked exceptions and gaps
Draft	Only supported factual claims; explicitly state missing answers
Cite and verify	Map each claim to its passage and check qualifiers again

Explain grounding and claim mapping

A **grounded answer** connects each factual claim to **evidence** actually retrieved for the task. Retrieval success does not automatically make generation faithful.

Build a claim-**evidence table** before final wording. Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table.

Term	Meaning
Grounded claim	A claim supported by the available task evidence
Qualifier	A condition restricting a statement
Evidence table	A mapping from each claim to its supporting record

Trace separate room C practice

Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch.

Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

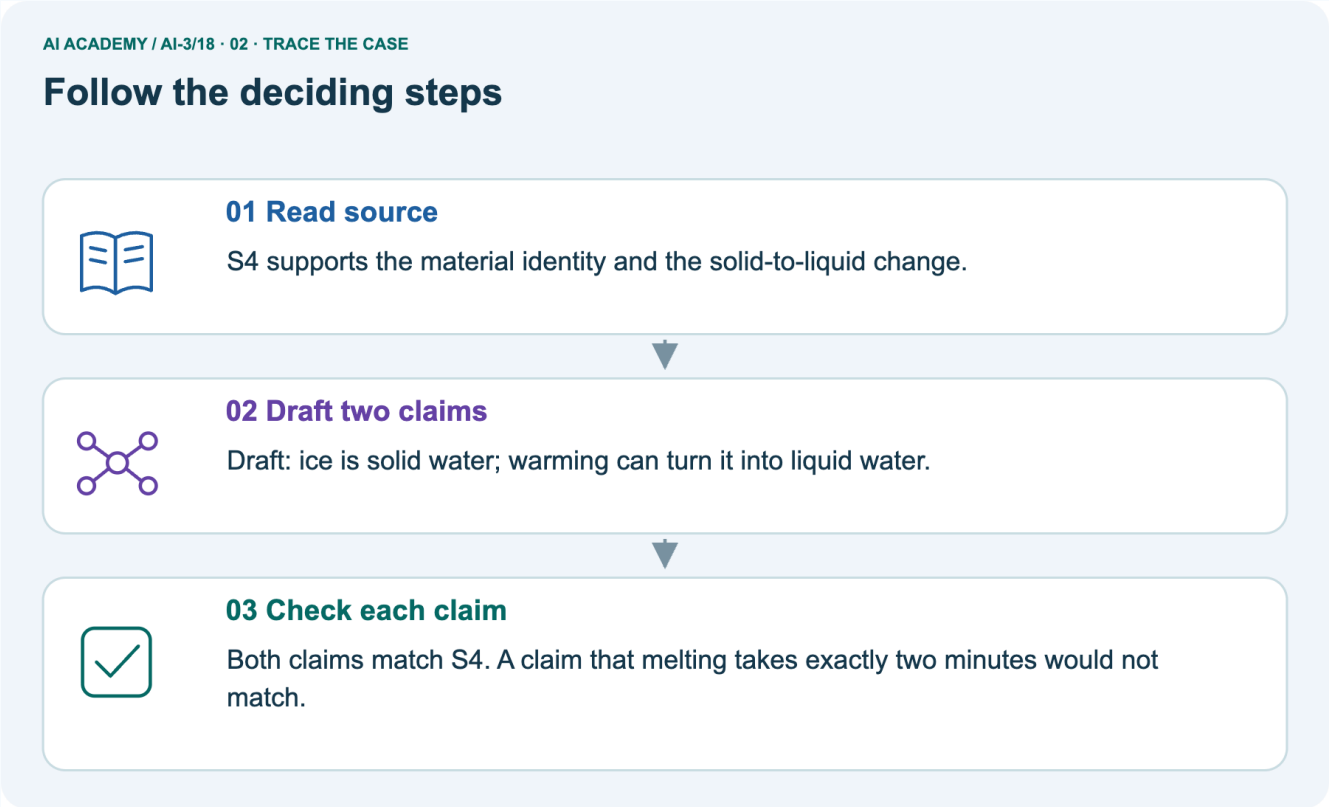
Read the complete S4 case

Turn a short science source into an answer with a visible claim-to-source map.

Approved note S4 says: ice is solid water; warming can melt ice into liquid water. No temperature or speed is supplied.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Map supported science claims



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read source	S4 supports the material identity and the solid-to-liquid change.
2. Draft two claims	Draft: ice is solid water; warming can turn it into liquid water.
3. Check each claim	Both claims match S4. A claim that melting takes exactly two minutes would not match.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Both claims match S4. A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict missing duration

Change: The question asks exactly how long this ice takes to melt.

Prediction: Say that S4 does not provide a time.

Reason: A general change of state does not determine a duration for an unspecified sample.

State source scope and picture limits

This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Extension · Chunk boundaries and exceptions

Chunking divides documents into retrievable units. A useful **chunk** retains enough structure to interpret a passage, including headers and exceptions.

Give each **chunk** its document/version ID, section and **offset**s. Preserve **qualifier**s or link adjacent material. **Overlap** can repeat information; repeated **chunk**s are not independent sources.

Guided: The first **chunk** alone lacks the applicable exception. Retrieve the linked condition and determine maintenance status before asserting capacity 20.

Qualification: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

▼ Extension · Toy lexical and dot-product ranks

Lexical retrieval matches terms; **vector retrieval** compares numerical representations. Each can fail when the representation misses the relevant distinction.

For a toy lexical score, count distinct **query** terms present after lowercase exact-word matching. For a toy vector score use a stated dot product. Compare **relevance** using independent judgments, not score magnitude alone.

Guided: Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task **relevance** favours the sensor manual; vector ranking alone does not prove **relevance**.

Qualification: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Question is converted into a search **query**.

2. **Retriever** returns four passages; selector keeps two relevant ones.

3. Generator answers using those passages and attaches citations.

4. If retrieval misses the needed **evidence**, generation should ask, search again, or abstain.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Apply the connected idea: Chunking and document structure

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

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Apply the connected idea: Lexical and vector retrieval

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Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

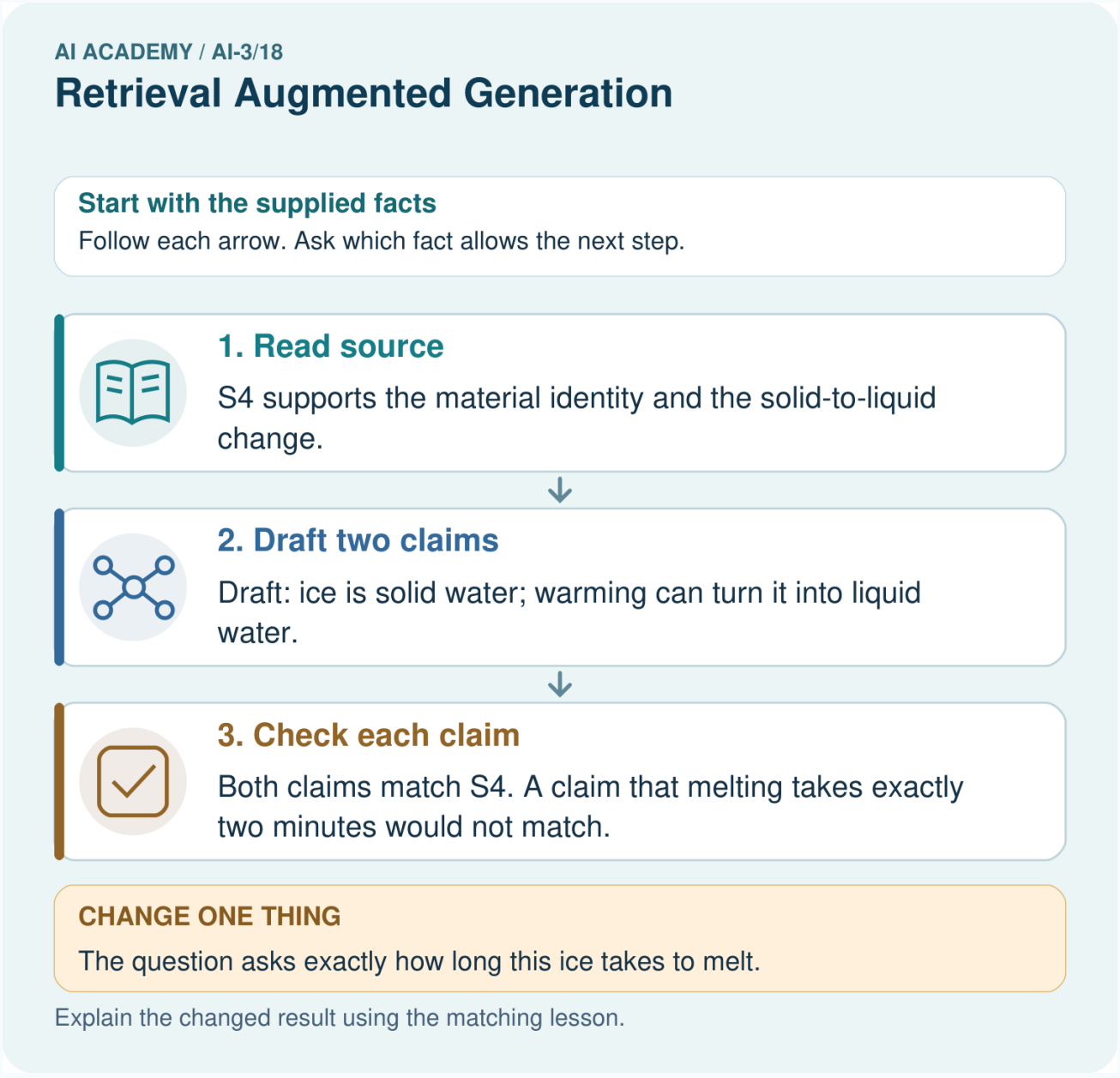
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Retrieval Augmented Generation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the main explanation, then fade support

> Collect each learner’s reasoning before revealing a key.

Main completion question and key

Completion task: Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Main individual check

Use the worked case for Building a **grounded answer**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed case

Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

None. Keep the supported state-change explanation and explicitly say the time is not supplied. Changed-case check: Say that S4 does not provide a time.

A general change of state does not determine a duration for an unspecified sample.

Changed-case reasoning: Say that S4 does not provide a time. A general change of state does not determine a duration for an unspecified sample.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep learners explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Extension teaching cycle

Target: Explain the **mechanism** and terminology of **chunk**ing and document structure

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: A notice says capacity 20, followed by during maintenance capacity 8. Separate **chunk**s contain each sentence but retrieval returns only the first.

Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The first **chunk** alone lacks the applicable exception. Retrieve the linked condition and determine maintenance status before asserting capacity 20.

Individual check: For **Chunk**ing and document structure, use its worked case to evaluate this proposed explanation: “The worked result establishes every future case”. Explain the deciding step.

Expected correction: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Connected teaching cycle: Lexical and vector retrieval

Target: Explain the **mechanism** and terminology of lexical and **vector retrieval**

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: **Query** terms red sensor. D1 text red sensor manual; D2 text red paint. Their lexical scores are 2 and 1. Vector **query** [1, 0], D1=[.4, .9], D2=[.8, .1].

Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task **relevance** favours the sensor manual; vector ranking alone does not prove **relevance**.

Individual check: For Lexical and **vector retrieval**, use its worked case to evaluate this proposed explanation: “The worked result establishes every future case”. Explain the deciding step.

Expected correction: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Note A says library opens at 9. Note B says returns go in the blue box.

Write the question to explain: Answer opening time with **evidence**.

Reveal after **prediction**s: Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

Use the earlier transfer question as optional practice. Reserve Ground the new room H response independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

Qualification: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.

Stage	Teacher action
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Teach the connected idea

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35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Connected teaching cycle: Chunking and document structure

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied main investigation

> Keep the selected independent task out of shared practice.

Supplied records and answer guidance

Paper RAG: S4-v1 alone supports ice identity and warming-related state change. R2 is unrelated; G3 is fiction, not a sample measurement; M1 is an empty template. Exact duration is unavailable. Room C review supports approval and capacity10, not lunch.

Independent room H supports approval and conditional limits7 normally/4 during maintenance, with status unknown; request the status rather than choose an unconditional capacity. Chunking extension preserves linked maintenance/Friday exceptions. Toy ranking

extension counts distinct exact query terms and uses the stated dot product; similarity is not relevance or truth.

Predict, investigate and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the four complete S4/R2/G3/M1 passage cards for paper RAG. Rank by the exact question, draft only supported claims, cite the source version and test the missing-duration question. Record paper traces without claiming a live retrieval or model run.
4. What did the hardest test teach you that the normal test did not?

Keep the evidence

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Main misconceptions

The worked result establishes every future case: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

The worked result establishes every future case: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Check these misconceptions

The worked result establishes every future case: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied main investigation for shared practice. Reserve Ground the new room H response independently for the individual first attempt. The named extension has separate tasks and feedback.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected main independent task

> Reveal its key after the first attempt; extension tasks have their own keys.

Selected independent question and key

Selected case: Source A approves room H; B lists H capacity 7, except maintenance limit 4. Maintenance status is unknown. Write a grounded response.

Teacher explanation: Room H approval is supported. State the conditional capacities 7 normally and 4 during maintenance, with status unresolved; do not choose an unconditional capacity.

Judge reasoning and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A **grounded answer** connects each factual claim to **evidence** actually retrieved for the task. Retrieval success does not automatically make generation faithful.

Build a claim-**evidence table** before final wording. Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table.

W2 Trace the worked case

Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

W3 Audit the limit

A report omits this qualification: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Extension questions and keys

Chunking and document structure — W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Chunk**ing divides documents into retrievable units. A useful **chunk** retains enough structure to interpret a passage, including headers and exceptions.

Give each **chunk** its document/version ID, section and **offset**s. Preserve **qualifier**s or link adjacent material. **Overlap** can repeat information; repeated **chunk**s are not independent sources.

Chunking and document structure — W2 Trace the worked case

A notice says capacity 20, followed by during maintenance capacity 8. Separate **chunk**s contain each sentence but retrieval returns only the first. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The first **chunk** alone lacks the applicable exception. Retrieve the linked condition and determine maintenance status before asserting capacity 20.

Chunking and document structure — W3 Audit the limit

A report omits this qualification: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

Chunking and document structure — W4 Apply a new case independently

A rule section says rooms close at 17:00; an exception paragraph says Friday closes at 16:00. The question is Friday closing. Design a complete **chunk** reference.

Teacher explanation: Return both rule and linked Friday exception with document/version and section references; answer 16:00 from the exception. An isolated 17:00 **chunk** is insufficient.

Chunking and document structure — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

The new case should test this particular **mechanism** rather than merely restate the original result.

Lexical and vector retrieval — W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Lexical retrieval matches terms; **vector retrieval** compares numerical representations. Each can fail when the representation misses the relevant distinction.

For a toy lexical score, count distinct **query** terms present after lowercase exact-word matching. For a toy vector score use a stated dot product. Compare **relevance** using independent judgments, not score magnitude alone.

Lexical and vector retrieval — W2 Trace the worked case

Query terms red sensor. D1 text red sensor manual; D2 text red paint. Their lexical scores are 2 and 1. Vector **query** [1, 0], D1=[.4, .9], D2=[.8, .1]. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task **relevance** favours the sensor manual; vector ranking alone does not prove **relevance**.

Lexical and vector retrieval — W3 Audit the limit

A report omits this qualification: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

Lexical and vector retrieval — W4 Apply a new case independently

Query terms blue battery. E1 text blue battery guide, vector [.2, .8]; E2 text blue ribbon, vector [.9, .1]. **Query** vector [1, 0]. Compute both rankings.

Teacher explanation: Lexical E1=2, E2=1; vector E1=.2, E2=.9 so rankings disagree. Check **relevance** for the battery task; do not equate dot product with source truth.

Lexical and vector retrieval — W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

Additional transfer practice

A retrieved passage says the club meets Tuesday; it says nothing about cost. The assistant answers Tuesday, free entry. Which part is grounded and what should replace the unsupported part?

Reasoned practice answer

Tuesday is supported. Free entry is unsupported; say the passage does not state the cost and seek an appropriate source if needed.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Retrieving one useful passage supports every detail in the final answer.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

It should say the supplied retrieved sources do not establish opening time and request better retrieval. The full collection contains A, so diagnose retrieval separately from generation.

A generator should not use cafe hours as library **evidence**.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Source A approves room H; B lists H capacity 7, except maintenance limit 4. Maintenance status is unknown. Write a grounded response.

Teacher explanation: Room H approval is supported. State the conditional capacities 7 normally and 4 during maintenance, with status unresolved; do not choose an unconditional capacity.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and return

> Keep main and extension assessment evidence distinguishable.

Project record

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Main support and fresh assessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Room H approval is supported. State the conditional capacities 7 normally and 4 during maintenance, with status unresolved; do not choose an unconditional capacity. Require intermediate reasoning and the stated scope.

Fresh retry: Source approves J; register says capacity 9 unless exam setup limits it to 6. Setup unknown.

Expected reasoning: Report approval and both capacity conditions; ask which setup applies before giving a single applicable limit.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Delayed return

Suggested return points in this level: ai-3/19, ai-3/21, ai-3/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 19: User, Need, and AI Contribution. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 19: User Need and AI Contribution. Carry forward source scope and evidence gaps when deciding whether AI serves the user's actual task.

▼ Extension support and reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Return both rule and linked Friday exception with document/version and section references; answer 16:00 from the exception. An isolated 17:00 **chunk** is insufficient. Require intermediate reasoning and the stated scope.

Fresh retry: General limit 30; exam-week exception 18. Question concerns exam week. Expected reasoning: Preserve both scope and exception; applicable limit 18. Two **overlap**ping **chunk**s do not count as two corroborating sources.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Readiness and reassessment for Lexical and vector retrieval

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Lexical E1=2, E2=1; vector E1=.2, E2=.9 so rankings disagree. Check **relevance** for the battery task; do not equate dot product with source truth. Require intermediate reasoning and the stated scope.

Fresh retry: **Query** green motor; F1 contains both, vector [.3, .7]; F2 green fabric, vector [.6, .4]; **query** [1, 0]. Expected reasoning: Lexical F1=2, F2=1; dot products .3, .6 favour F2.

Use **relevance judgment**s to evaluate the representation. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Implement question → retrieve → inspect → answer → cite → verify as the core architecture.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Bridge to the next week

Support and fresh reassessment

Readiness and reassessment for Chunking and document structure

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative teaching checklist**

REFERENCE · SECTION 7

Reference · Alternative teaching checklist

> Use as a reminder alongside the main route.

Planning reference

Teaching target: Explain the **mechanism** and terminology of building a **grounded answer**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

Check every learner before discussion: Use the worked case for Building a **grounded answer**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The question asks exactly how long this ice takes to melt. Require a **prediction** and a reason before comparing. Expected result: Say that S4 does not provide a time.

Reason: A general change of state does not determine a duration for an unspecified sample.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and retained evidence**

REFERENCE · SECTION 8

Reference · Feedback and retained evidence

> Keep supported practice separate from independent evidence.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

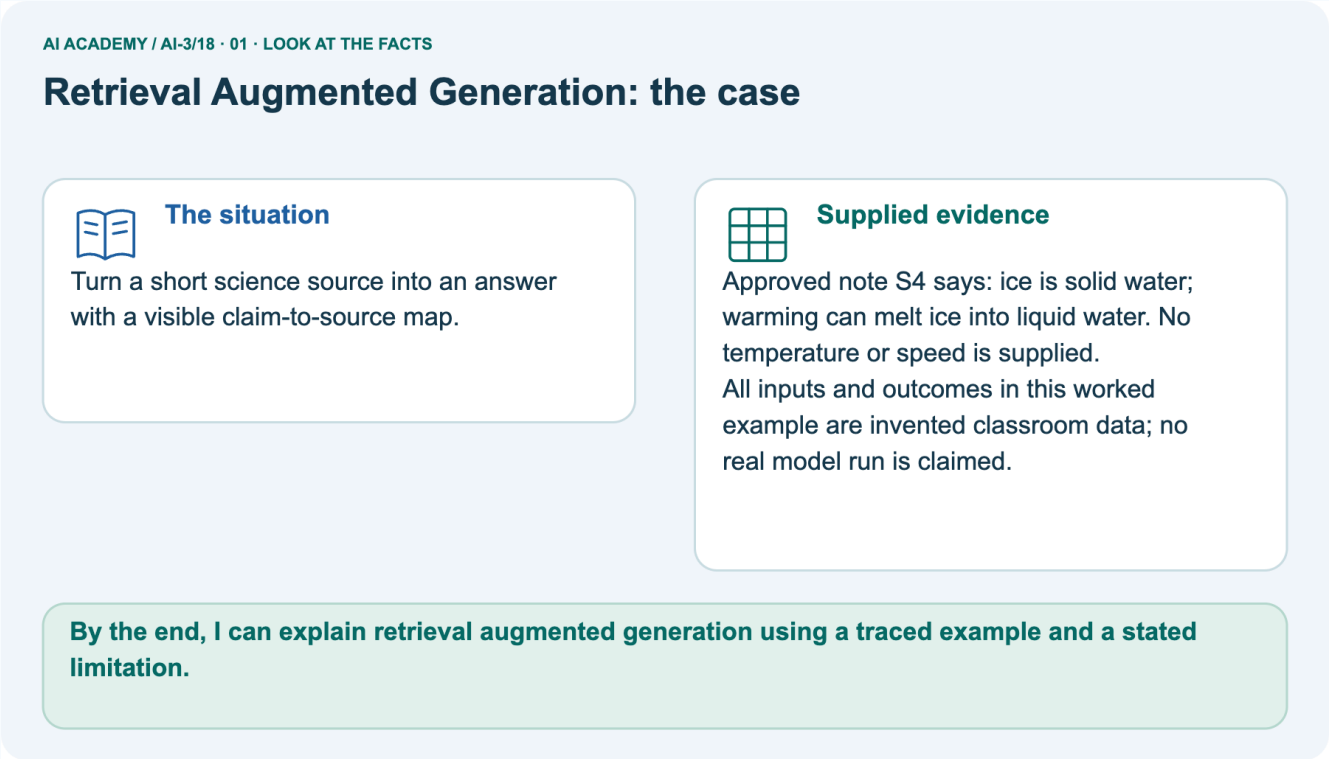
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose a picture for a specific explanation.

01. Retrieval Augmented Generation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

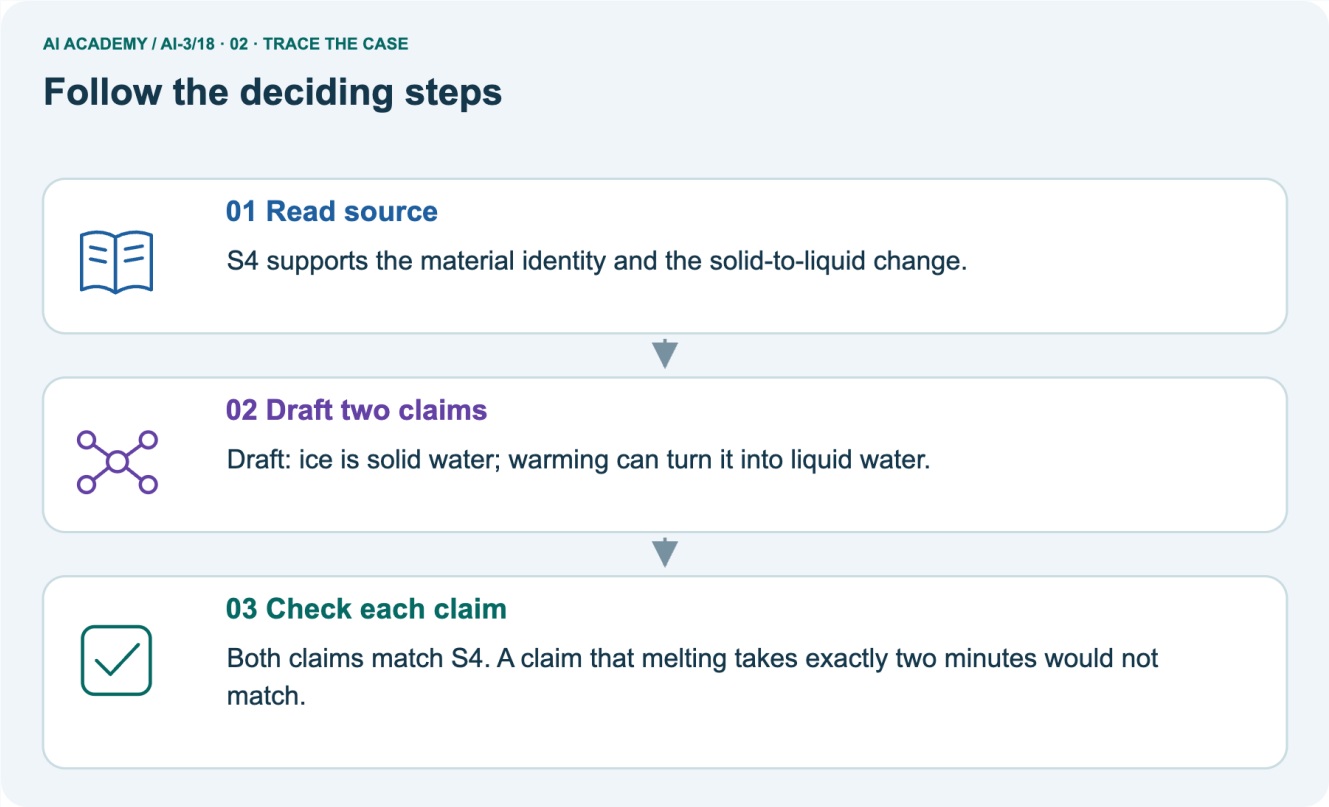
Turn a short science source into an answer with a visible claim-to-source map. Supplied facts: Approved note S4 says: ice is solid water; warming can melt ice into liquid water. No temperature or speed is supplied.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

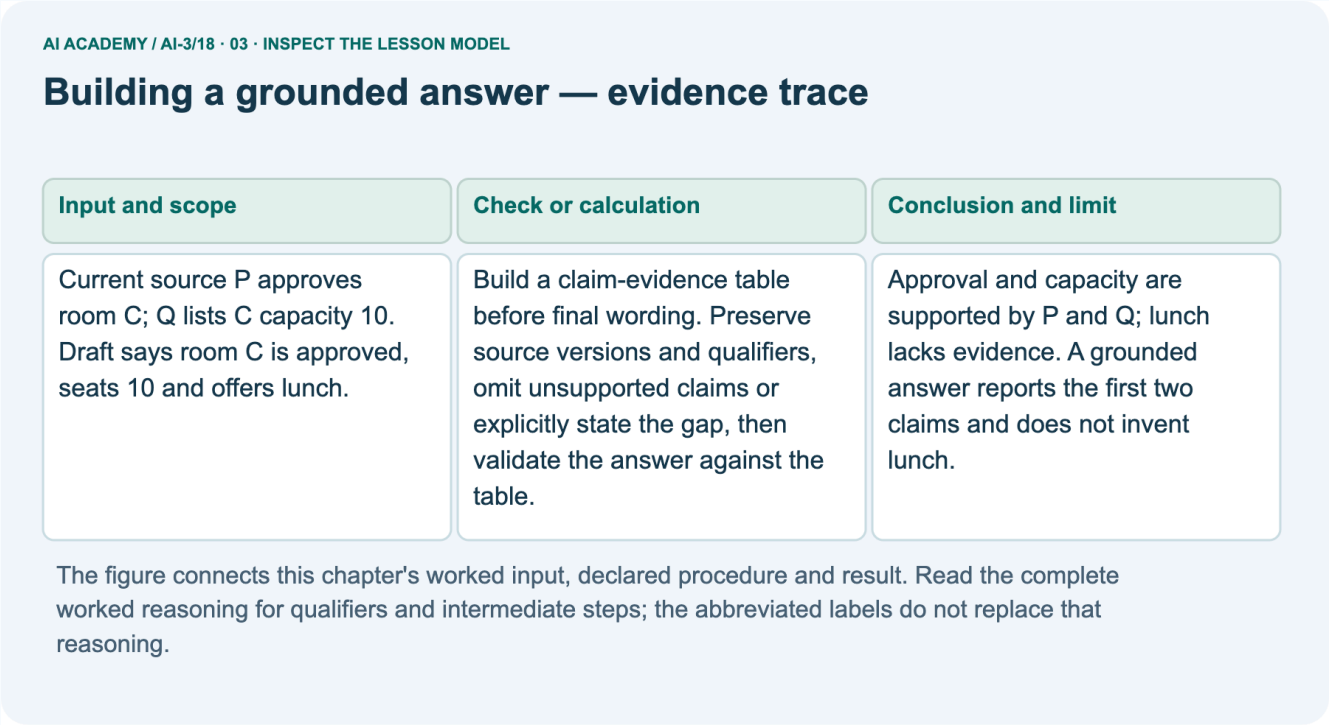
Read source: S4 supports the material identity and the solid-to-liquid change. → Draft two claims: Draft: ice is solid water; warming can turn it into liquid water. → Check each claim: Both claims match S4.

A claim that melting takes exactly two minutes would not match.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Building a grounded answer — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Building a **grounded answer** — **evidence** trace. Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. | Build a claim-**evidence table** before final wording.

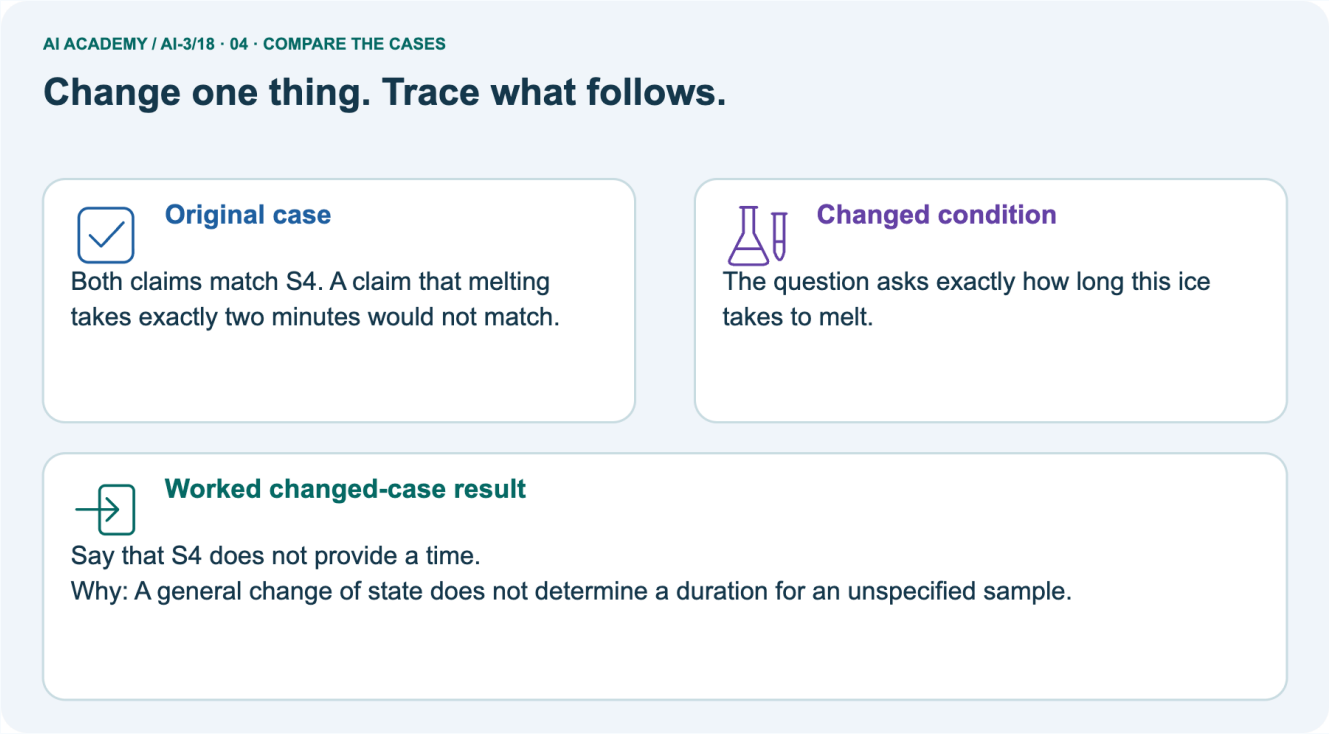
Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table. | Approval and capacity are supported by P and Q; lunch lacks **evidence**.

A **grounded answer** reports the first two claims and does not invent lunch.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both claims match S4. A claim that melting takes exactly two minutes would not match. Changed condition: The question asks exactly how long this ice takes to melt.

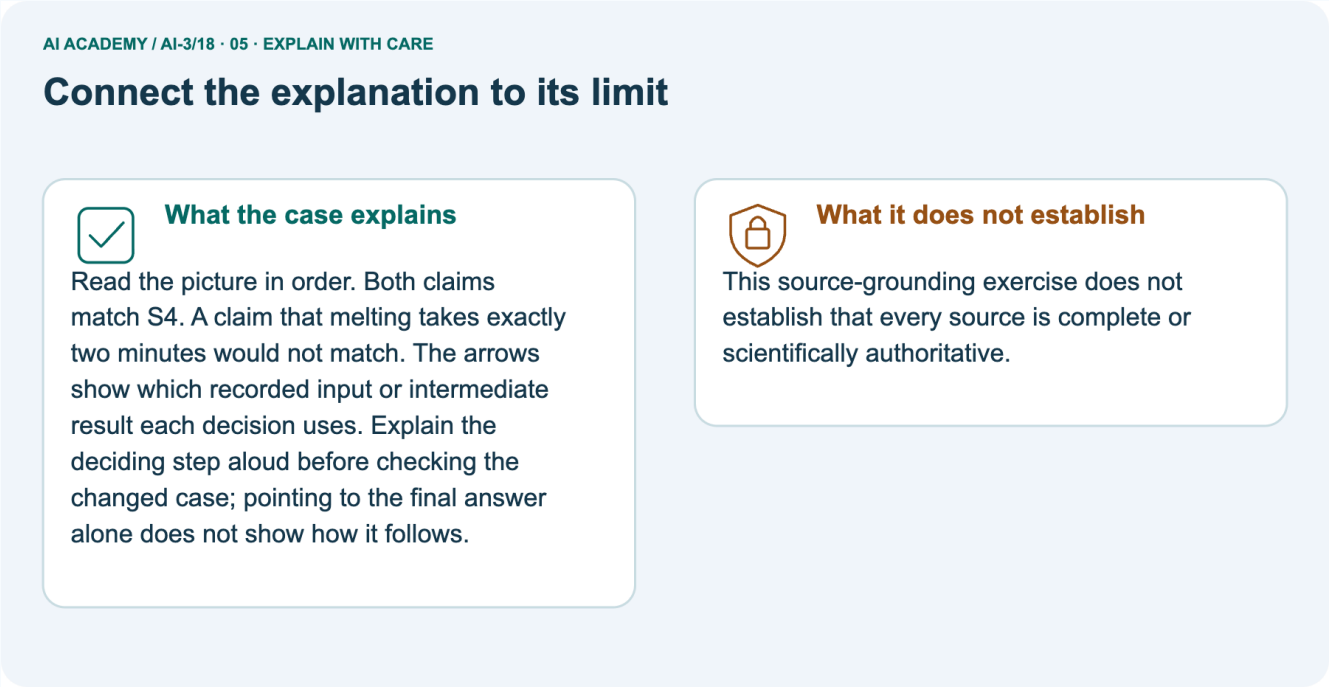
Expected result: Say that S4 does not provide a time. Reason: A general change of state does not determine a duration for an unspecified sample.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A general change of state does not determine a duration for an unspecified sample.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Both claims match S4. A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

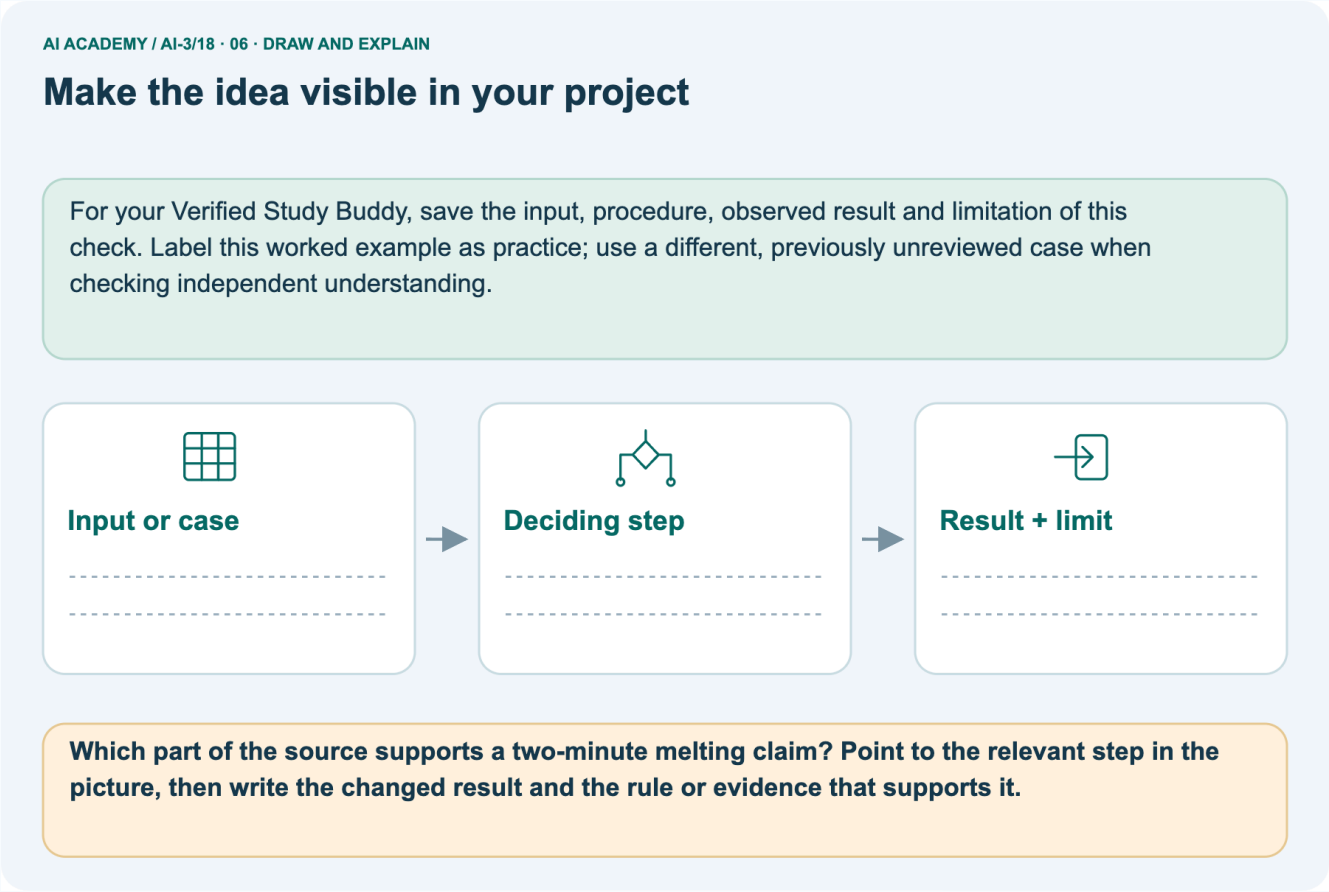
Limit: This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: None. Keep the supported state-change explanation and explicitly say the time is not supplied. Changed-case check: Say that S4 does not provide a time.

A general change of state does not determine a duration for an unspecified sample.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Retrieval Augmented Generation: the case

02. Follow the deciding steps

03. Building a grounded answer — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.